

OS Practical No. 9

Aim :

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Optimal page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:

```
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3];
    int n = 12, frames_count = 3;
    int i, j, k, pos, farthest;
    int hit = 0, fault = 0;
    int flag;

    for(i = 0; i < frames_count; i++)
        frames[i] = -1;

    for(i = 0; i < n; i++) {
        flag = 0;

        // Check for page hit
        for(j = 0; j < frames_count; j++) {
            if(frames[j] == pages[i]) {
                hit++;
                flag = 1;
                break;
            }
        }

        // Page fault
        if(flag == 0) {
            pos = -1;
            farthest = i + 1;
        }
    }
}
```

```

    for(j = 0; j < frames_count; j++) {
        int found = 0;
        for(k = i + 1; k < n; k++) {
            if(frames[j] == pages[k]) {
                if(k > farthest) {
                    farthest = k;
                    pos = j;
                }
                found = 1;
                break;
            }
        }

        if(found == 0) {
            pos = j;
            break;
        }
    }

    if(pos == -1)
        pos = 0;

    frames[pos] = pages[i];
    fault++;
}

float hit_percent = (hit * 100.0) / n;
float fault_percent = (fault * 100.0) / n;

printf("Total Page Hits: %d\n", hit);
printf("Total Page Faults: %d\n", fault);
printf("Page Hit Percentage: %.2f%%\n", hit_percent);
printf("Page Fault Percentage: %.2f%%\n", fault_percent);

return 0;
}

```

Output :

```
UW PICO 5.09
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3];
    int n = 12, frames_count = 3;
    int i, j, k, pos, farthest;
    int hit = 0, fault = 0;
    int flag;

    for(i = 0; i < frames_count; i++)
        frames[i] = -1;

    for(i = 0; i < n; i++) {
        flag = 0;

        // Check for page hit
        for(j = 0; j < frames_count; j++) {
            if(frames[j] == pages[i]) {
                hit++;
                flag = 1;
                break;
            }
        }

        // Page fault
        if(flag == 0) {
            pos = -1;
            farthest = i + 1;

            for(j = 0; j < frames_count; j++) {
                int found = 0;
                for(k = i + 1; k < n; k++) {
                    if(frames[j] == pages[k]) {
                        if(k > farthest) {
                            farthest = k;
                            pos = j;
                        }
                    }
                    found = 1;
                    break;
                }

                if(found == 0) {
                    pos = j;
                    break;
                }
            }

            if(pos == -1)
                pos = 0;

            frames[pos] = pages[i];
            fault++;
        }
    }

    ^G Get Help      ^O WriteOut      ^R Read File
    ^X Exit          ^J Justify       ^W Where is
```

```
[saumyaudgirkar@Saumyas-MacBook-Air ~ % gcc optimal.c -o optimal
[saumyaudgirkar@Saumyas-MacBook-Air ~ % ./optimal
Total Page Hits: 5
Total Page Faults: 7
Page Hit Percentage: 41.67%
Page Fault Percentage: 58.33%
saumyaudgirkar@Saumyas-MacBook-Air ~ % █
```